Lab Task

Subject: **Operating System**

Topic: System call and file handling

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Submitted To:

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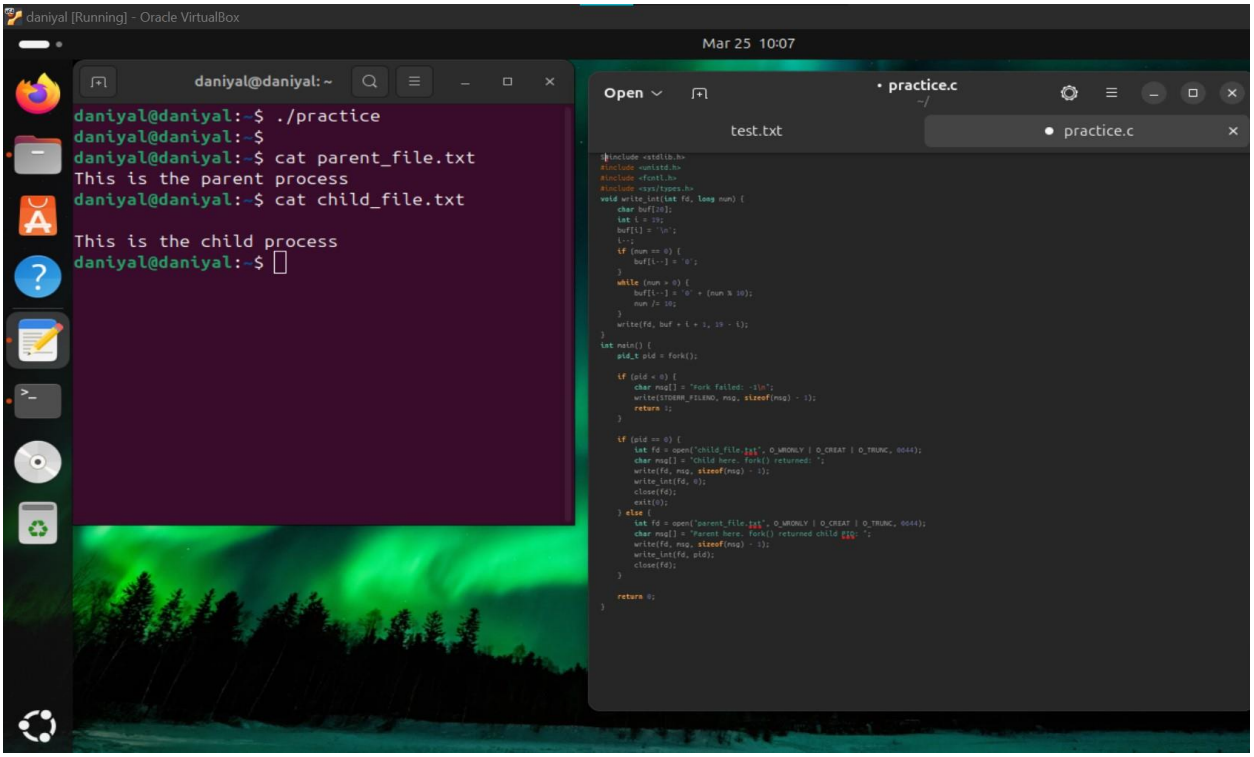
PART # 1

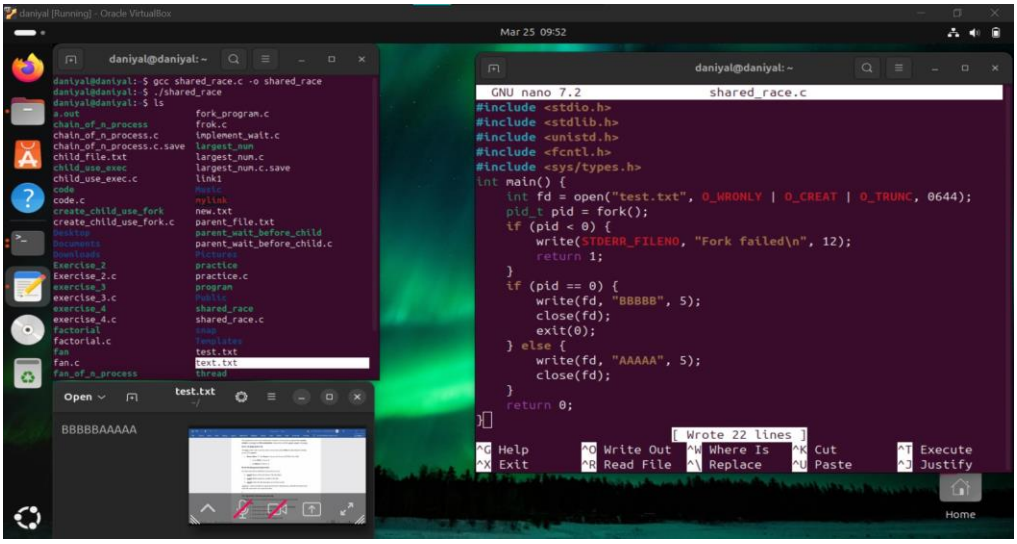


CODE

```
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/types.h>
void write_int(int fd, long num) {
   char buf[20];
   int i = 19;
   buf[i] = '\n';
   i--;
   if (num == 0) {
      buf[i--] = '0';
   }
   while (num > 0) {
      buf[i--] = '0' + (num % 10);
      num /= 10;
   }
   write(fd, buf + i + 1, 19 - i);
}
int main() {
   pid_t pid = fork();

   if (pid < 0) {
      char msg[] = "Fork failed: -1\n";
      write(STDERR_FILENO, msg, sizeof(msg) - 1);
      return 1;
   }

   if (pid == 0) {
      int fd = open("child_file.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```
    char msg[] = "Child here. fork() returned: ";
    write(fd, msg, sizeof(msg) - 1);
    write_int(fd, 0);
    close(fd);
    exit(0);
  } else {
    int fd = open("parent_file.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    char msg[] = "Parent here. fork() returned child PID: ";
    write(fd, msg, sizeof(msg) - 1);
    write_int(fd, pid);
    close(fd);
  }

  return 0;
}
```

PART # 2



CODE

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <fcntl.h>

#include <sys/types.h>

int main() {

  int fd = open("test.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

  pid_t pid = fork();

  if (pid < 0) {

    write(STDERR_FILENO, "Fork failed\n", 12);

    return 1;

  }
```

```
    if (pid == 0) {
        write(fd, "BBBBB", 5);
        close(fd);
        exit(0);
    } else {
        write(fd, "AAAAA", 5);
        close(fd);
    }
    return 0;
}
```